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Neuralite: Enabling Wireless High-Resolution Brain-Computer Interfaces

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Abstract

Intracortical brain-computer interfaces (iBCIs) promise to sense brain activity at an unprecedented scale and resolution. However, unlocking this potential for practical, untethered applications remains an unsolved challenge. The major barrier is the significant wireless bandwidth required to stream high-resolution brain signals. Existing approaches rely on extensive on-device processing, which is severely constrained by the limited resources of iBCI devices, the complexity of brain signals, and the dynamic nature of neural activity. This paper introduces *Neuralite*, a wireless iBCI system that integrates high-fidelity brain signal models and effective brain sensing mechanisms within an efficient server-driven streaming framework. By thoroughly characterizing brain signal variability, Neuralite adaptively optimizes streaming under dynamic neural conditions, minimizing bandwidth consumption without imposing excessive burdens on resource-constrained iBCI devices. Experimental results demonstrate that Neuralite significantly reduces bandwidth consumption while preserving neural decoding precision across key iBCI components and representative applications.

CCS Concepts

• **Human-centered computing → Ubiquitous and mobile computing systems and tools**; • **Networks → Network protocol design**.

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Keywords

Brain-computer interface, Spike sorting, Neural decoding

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1 Introduction

Recent years have witnessed a mounting interest in brain-computer interfaces (BCIs). Among the various types [14, 35], intracortical BCIs (iBCIs) offer unparalleled signal resolution, enabling a wide range of applications including psychology and neuroscience research [25, 47, 69], treatments of epilepsy [7, 46] and Parkinson's disease [27, 75], and advanced prosthetics that restore motor-sensory functions lost to disabilities [2, 5, 11, 34, 40, 50, 72, 73]. Building on recent advancements in low-invasive brain implants [67], next-generation BCIs promise a generalized brain interface that merges machine intelligence and the human mind [55].

A typical iBCI system comprises an implanted microelectrode array (MEA), a small wearable device called a headstage, and a neural decoder offloaded to an edge server. The MEA records brain signals and transmits them to the headstage via thin wires. The headstage amplifies and digitizes these signals before sending them to the neural decoder, which interprets brain activity using deep neural networks (DNNs). Similar to how high-definition cameras capture images with a large pixel array, an MEA consists of densely packed microelectrodes, each monitoring a small group of nearby neurons. Since neurons exchange information by firing spikes, the first step in neural decoding is to isolate the signals from individual neurons through *spike sorting* [12], a process that utilizes unsupervised learning to differentiate spikes based on the unique waveforms shaped by each neuron's biophysical properties. By collectively recording a large neuronal population while maintaining single-neuron resolution through spike sorting, state-of-the-art iBCIs have demonstrated unprecedented precision in decoding complex brain activities.

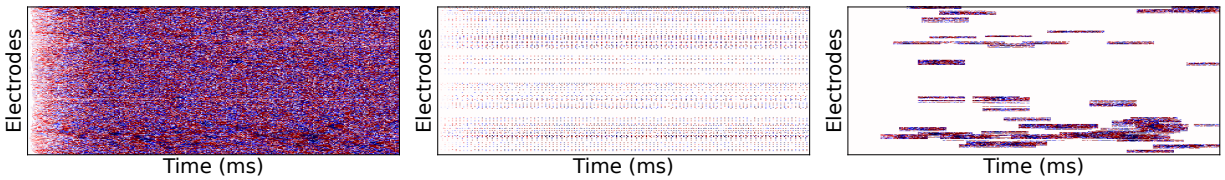

Figure 1: Server-driven streaming of an 100 ms brain signal segment recorded with a 374-electrode MEA. From left to right: the full-resolution frame (17.9 Mb). The basic frame recorded by the core electrode set (0.94 Mb). The enhancement frame constructed based on streaming feedback (1.74 Mb).

Despite its immense potential, realizing practical, untethered iBCIs remains an unsolved challenge. The primary barrier is the significant wireless bandwidth required to stream high-resolution (HR) brain signals. On one hand, iBCI headstages are severely constrained by power budgets and processing capacity, restricting their transmission rate to just tens of Mbps [37]. On the other hand, modern MEAs contain hundreds to thousands of electrodes [26, 36, 64], which typically require hundreds of Mbps in bandwidth. Recent studies show that MEAs' electrode density has been steadily increasing following Moore's law [65]. While this trend promises a significant boost in neural decoding precision, it also leads to a proportional rise in bandwidth demands, exacerbating the already critical bandwidth bottleneck.

To overcome the bandwidth limitation, previous untethered iBCIs have relied on the headstage to compress brain signals [20, 31, 58, 59, 70, 74, 76, 80], extract signal features [1, 15, 24, 29, 57], or decode brain activity directly [28, 63]. These *headstage-based* methods encounter two fundamental challenges. *First*, iBCI headstages are limited by power, size, and weight, severely restricting their on-device resources and processing capabilities. For instance, to ensure safety and stability, the Neuralink headstage [36] measures only 23mm × 18.5mm × 2mm and weighs 15g, making it 20.4x smaller and 2.4x lighter than an Apple Watch SE. Moreover, because headstages are in close contact with the scalp, excessive heat generated from intensive signal processing can affect neural activity, potentially degrading recording fidelity. *Second*, HR brain signals present significant complexities. Due to the dense arrangement of recording electrodes, the spatial features of spikes are often heavily intertwined across electrodes. Moreover, when nearby neurons fire simultaneously, their spike waveforms can overlap on the same electrode, resulting in neural interference that corrupts spike features, reduces sorting accuracy, and degrades neural decoding performance. Due to stringent resource constraints of the headstage, on-device processing is inherently ineffective at resovling these complexities.

This paper addresses the following question: *Can we untether iBCIs under limited wireless bandwidth without excessive brain signal processing on resource-constrained headstages?* To generalize our study across diverse downstream applications, we focus on spike sorting, the foundational building block for the majority of neural decoding pipelines. Rather than relying on the headstage for extensive brain signal processing, our key idea is to shift streaming control to the edge server. To validate this strategy, we conduct measurements on a diverse set of brain signal datasets recorded from major brain regions. Our results show that, although spike power often spreads widely across a large number of electrodes, the critical features are typically concentrated on a much smaller electrode subset. Leveraging this insight, we design a brain signal streaming system that follows a *server-driven* workflow, as depicted in Fig. 1. During initialization, the headstage transmits a small amount of full-resolution signals, allowing the server to identify a core electrode set that can reliably detect spikes from all neurons. For each subsequent signal segment, the headstage first sends a *basic frame* recorded by the core electrode set. The server then analyzes the basic frame to derive streaming feedback, guiding the headstage in constructing a small *enhancement frame* containing key features for accurate spike sorting.

Our server-driven design offers three key advantages. First, offloading streaming control to the server significantly reduces the processing burden on headstages. Second, by utilizing computational resources at the server for in-depth characterization of brain signal complexity, our system can derive optimal streaming feedback, minimizing bandwidth consumption without compromising spike sorting accuracy. Third, server-driven streaming operates independently of compression schemes. When adequate resources are available, the headstage can further compress the basic and enhancement frames to save additional bandwidth.

While recent works have explored a similar approach in machine-centric video systems [9, 32, 78, 79], server-driven streaming of HR brain signals presents unique challenges. First, neural interference leads to significant signal variability, making it difficult to identify a core electrode set for reliable spike detection. Second, the spatial features of HR brain signals are highly intertwined, requiring careful feedback optimization to balance bandwidth efficiency with spike sorting accuracy. Third, to implement server-driven streaming, the headstage must buffer brain signals during feedback intervals and construct frames in real-time. Given the limited speed and capacity of headstage memory, these operations

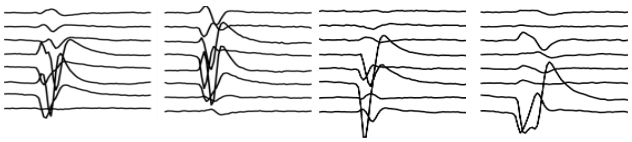

Figure 2: Spike templates of four neurons recorded by eight electrodes.

can introduce substantial overhead, potentially degrading streaming efficiency.

We present *Neuralite*, a server-driven brain signal streaming system for wireless iBCIs. To address the uncertainties introduced by stochastic neural interference, we develop a faithful neural activity simulator that can generate realistic interference scenarios from minimal training signals. By extensively profiling brain signal variability through high-fidelity simulation, our stream controller accurately characterizes spike power and feature distributions across electrodes. We then design an efficient greedy heuristic to construct the core electrode set based on spike power distribution, minimizing basic frame size while ensuring reliable spike detection. To optimize streaming feedback, we propose a saliency-based mechanism to identify the subset of electrodes that captures key spike features. We then further refine this subset based on the neural interference model, enabling Neuralite to adapt enhancement frames under dynamic neural interference conditions.

We implement Neuralite headstage using a low-power dual-core microcontroller unit (MCU) [66]. To overcome the bottleneck of slow memory, we orchestrate the dual cores to construct and transmit brain signal frames in parallel. To further boost processing throughput, the headstage incorporates a transmission pipeline that harnesses the feedback latency. To reduce storage costs, we develop a buffer controller that exploits the bursty nature of neural activity to opportunistically skip unnecessary brain signals. We prototype our headstage on a customized hardware measuring 26.9mm × 21mm × 2mm and weighing 4.64g excluding battery, which are comparable to the most compact commodity headstages and research prototypes.

We evaluate Neuralite by using our headstage prototype to replay brain signal datasets recorded from major brain regions. We compare Neuralite against seven representative headstage-driven schemes, assessing their impact on two state-of-the-art spike sorting algorithms [44, 77], as well as four neural decoders that infer the motion of a rat [13, 56], the imagined handwriting of a paralyzed patient [73], and the visual scenes observed by mice [56, 71]. Our results show that: (1) Across all datasets, Neuralite limits spike sorting error to below 4.6%, reducing the error by 2.2x to 13.9x compared to the best headstage-driven scheme; (2) Neuralite consistently outperforms the headstage-driven schemes in a series of microbenchmarks across various wireless bandwidth, neuronal activity intensities, electrode densities, and MEA sizes; (3) At the application level, Neuralite limits the increase in neural decoding errors to between 1.3% and 1.7%, which is 17x to 31x lower than that of headstage-driven schemes.

2 Background and Related Work

Spike sorting. By isolating and identifying the spikes of individual neurons, spike sorting allows downstream applications to track brain activity at single-neuron resolution, resulting in significant enhancement of neural decoding precision [68]. Current spike sorting algorithms generally fall into two categories. *Cluster-based algorithms* [52, 53] group spikes from the same neuron by analyzing waveform features, such as principal components, wavelet coefficients, and shape characteristics. In contrast, *template-based algorithms* [43–45, 77] assign spikes to neurons by matching the spike waveform to *spike templates*. Specifically, the spike template of a neuron is the average spike waveforms recorded across the MEA. For instance, Fig. 2 shows the spike templates of four neurons recorded by eight electrodes. Typically, template-based algorithms establish spike templates during a training phase using a cluster-based method. However, due to implant drift and gradual changes in the MEA's electrical properties, spike templates may vary slowly over time. To track these variations, spike sorting algorithms typically analyze residual sorting errors and update the spike templates periodically. For instance, Kilosort3 [44] updates spike templates every 2.2 seconds.

Wireless iBCIs. To overcome bandwidth limitations, existing wireless iBCIs rely on wideband radios that support high data rates [61]. However, wideband radios are power-hungry, leading to short battery life and the risk of overheating, which can interfere with neural activity. To alleviate bandwidth demands, prior studies have applied wavelet transformation to compress brain signals [20, 58, 59]. Unfortunately, traditional compression methods fail to account for brain signal complexity, causing severe waveform distortion and significant degradation in neural decoding performance. To preserve spike features during compression, recent works have explored compressive sensing [76, 80], autoencoding [70, 74], and neuron-aware wavelet transformation [31] to optimize compression based on spike templates learned during the training phase. However, these methods remain susceptible to dynamic neural interference and impose excessive burdens on headstages. To address these limitations, Neuralink [36] bypasses raw signal transmission altogether and sends only basic spike statistics, such as firing rates. Because it does not support spike sorting, its applications are limited to simple peripherals with a low degree of freedom, making it inadequate for advanced prosthetics.

Dataset	MEA	State	Brain region
Human-1 and Human-2 [47]	374-electrode Neuropixel	Sober	Temporal and frontal lobes
Rat [33]	374-electrode Neuropixel	Anesthesia	Visual cortex, hippocampus
Mice [77]	256-electrode Utah array	Sober	Ganglion cells
Macaque [6]	96-electrode Utah array	Sober	Motor cortex

Table 1: Summary of brain signal datasets.

On-chip iBCIs. On-device inference has received significant attention. Various approaches have been proposed to adapt deep learning frameworks for resource-constrained IoT devices [16–19]. However, existing solutions fail to account for brain signal complexity and do not meet the more stringent resource constraints of headstages. Recently, several purpose-built on-chip systems have been developed for untethered iBCIs [1, 28, 29, 57, 63]. To handle the high data rates of HR brain signals, these systems typically simplify spike sorting by performing template matching on a single selected electrode using a compressed template, which inevitably reduces sorting performance. In contrast, state-of-the-art spike sorting algorithms rely on dense recordings from multiple electrodes to achieve high resolution, and require multiple rounds of iterative template matching to resolve neural interference [43–45]. As electrode density continues to increase following Moore's law, the processing burden will rise proportionally, presenting a fundamental challenge for on-device iBCIs.

3 Characterizing HR Brain Sensing

This section characterizes HR brain sensing with a focus on spike power and feature distributions across the MEA. Our angle contrasts with neuroscience studies that model individual spike waveforms and firing processes [45, 48, 51, 54]. In addition to motivating the server-driven design of Neuralite, our findings can also inform future iBCI system research, such as optimizing iBCI power consumption by adapting electrode parameters based on the characteristics of surrounding neurons [10].

Methodology. We conduct measurements on five brain signal datasets recorded from major brain regions of human, rat, mice, and macaque, as summarized in Table 1. We sort spikes using Kilosort-3 [44], a template-based algorithm widely used by the neuroscience community. Once training is done, we analyze spike power and feature distributions by constructing two bipartite graph models.

• *Power graph.* Consider the spike template of a neuron n expressed as a matrix W_n, where the e-th vector $\mathbf{w}_{n,e}$ represents the average spike waveform of neuron n recorded by electrode e. Following the widely adopted signal detection criteria, we consider the spike power of neuron n as detectable by electrode e if the peak SNR of $\mathbf{w}_{n,e}$ exceeds 3 dB [49, 62]. We then construct a bipartite power graph by linking electrodes and their detectable neurons. Fig. 3 (left) shows the adjacency matrix of the power graph of the Human-1 dataset.

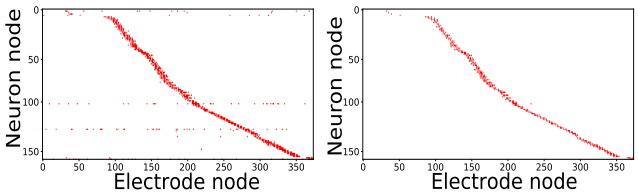

Figure 3: The adjacency matrices of power (left) and feature graph (right) of the Human-1 dataset.

• *Feature graph.* To measure spike feature distributions, we apply Singular Value Decomposition (SVD) to interpret spike templates. The SVD of the spike template of a neuron n can be expressed as $W_n = U_n \Sigma V_n$, where U_n denotes the spatial feature matrix of W_n, in which the i-th column $\mathbf{u}_{n,i}$ represents the i-th principal component (PC) and the value of $u_{n,i,e}$ reflects the contribution of electrode e to that PC. Empirically, we observe that spike features are concentrated on the first three PCs, which is consistent with prior spike sorting studies [45]. To quantify the contribution of electrode e in sorting the spike of neuron n, we compute a *feature value* as $\varphi_{n,e} = u_{n,0,e} + u_{n,1,e} + u_{n,2,e}$. Similar to constructing the power graph, we build a bipartite feature graph by linking neurons and electrodes where the feature values are 3 dB higher than the noise floor. Fig. 3 (right) shows the adjacency matrix of the feature graph for the human dataset.

Results. We quantify spike power and feature distributions by measuring the vertex degrees in constructed power and feature graphs. To characterize the impacts of neural interference, we measure the variability of spike power and feature across individual spikes fired by the same neuron.

• *Spike power distribution.* To characterize spike power distribution, we measure the degrees of electrode and neuron vertices in power graphs. As plotted in Fig. 4, we observe high degrees on a majority of neuron vertices across all studied datasets, reflecting the wide spread of spike power across densely packed electrodes. As another result of this power spread, we also observe high degrees on electrode vertices. For instance, in the rat dataset recorded by the 374-electrode Neuropixel, the median degree of electrode vertices is 8, indicating 50% electrode can detect more than 8 neurons. Such high degree of electrode vertices is particularly apparent in datasets recorded by MEAs of higher densities. For example, in the Human dataset recorded using a 374-electrode Neuropixel, the average electrode degree is 6.4, which is

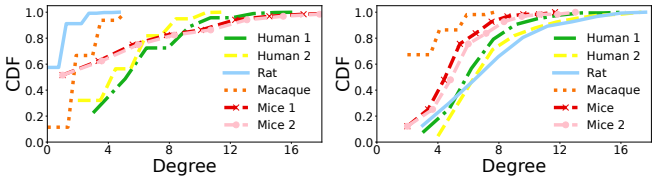

Figure 4: The degrees of electrodes (left) and neurons (right) in the power graph.

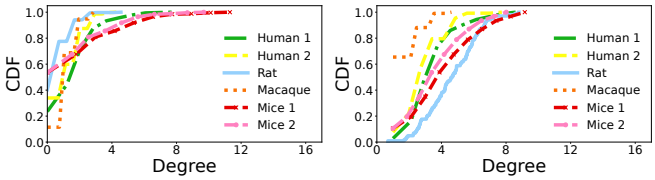

Figure 5: The degrees of electrodes (left) and neurons (right) in the feature graph.

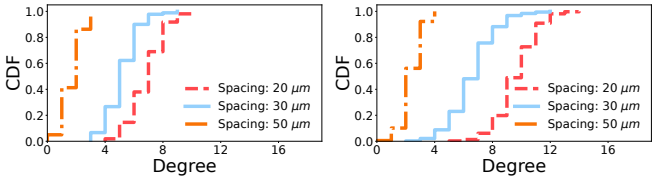

Figure 6: Electrode degrees in power (left) and feature (right) graphs v.s. MEA density.

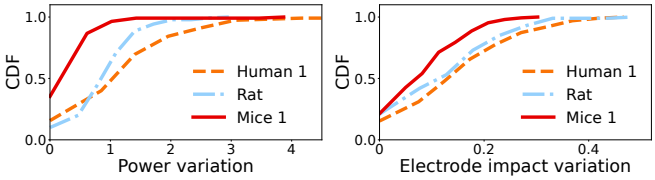

Figure 7: Spike power and feature variations in the presence of neural interference.

4x higher than that in the Macaque dataset recorded by a 96-electrode Utah array.

• *Spike feature distribution.* Fig. 5 plots the degrees of electrode and neuron vertices in feature graphs. Similar with the results in power graphs, we observe that the degrees of both types of vertices increase with MEA density. However, compared with spike power distribution, spike features are more concentrated on a smaller number of electrodes. For example, in the Human dataset recorded by Neuropixel, the average degree of neuron vertices is only 4.4, which is 1.45x smaller than that in the power graph of the same dataset.

• *Generalizability.* Our measurements show that, although spike power can spread widely across many electrodes in an MEA, the relevant features are often concentrated on a smaller subset of electrodes. To validate the generalizability of this result across MEAs with different densities, we examine spike power and feature distributions as a function of electrode spacing. Fig. 6 shows the distributions of electrode degrees in power and feature graphs for MEAs with different electrode spacings, including a 374-electrode Neuropixel, a 256-electrode MEA, and a 96-electrode Utah array, with electrode spacings of 20 μm, 30 μm, and 50μm, respectively. As can be observed in Fig. 6, when electrode spacing decreases from 30 μm to 20 μm, the median power graph degree increases by 2x, whereas the feature graph degree increases by only 1.2x. Meanwhile, the difference between power and feature graph degrees increases by 3x. This can be attributed to two factors: (1) as MEA density increases, spike powers spread across more electrodes; and (2) densely packed electrodes often record similar spike waveforms with overlapping features, resulting in a much slower increase in feature graph degree.

• *Impacts of neural interference.* Since the power and feature graphs studied in Fig. 4 and 5 are constructed based on spike templates, they do not reflect the impact of dynamic neural interference. To measure this impact on spike power distribution, we compute the ratio between the powers of each recorded spike and the spike templates of the same neuron. Fig. 7 (left) plots the cumulative distribution of this power ratio. Due to neural interference, we observe significant spike power variations, especially in datasets recorded by MEAs with higher densities. For example, in the Human dataset recorded by a 374-electrode Neuropixel probe, the power ratio can reach up to 5.2.

To characterize how neural interference affects spike feature distribution, we first estimate the influence of electrodes on spike sorting performance, and then measure how this influence varies across individual spikes fired by the same neuron. Specifically, given an electrode e and a neuron n, we compute an *influence score* defined as the ratio of e's feature value $\varphi_{n,e}$ to the feature value sum of neuron n. Fig. 7 shows the cumulative distribution of influence scores. For each neuron, we focus on the electrode that has the maximum influence score for that neuron's spike template. Similar with the results of power variation, we observe significant variation in influence scores. For example, in the Human-1 dataset, the influence score of an electrode can vary by up to 42%.

Summary. Our measurement reveal several key insights. (1) The spike power of HR brain signals is often widely distributed across a large number of electrodes, presenting an opportunity to construct a small subset of electrodes capable of detecting spikes from all neurons; (2) In contrast to spike power distribution, spike features tend to concentrate on a much smaller number of electrodes, enabling a streaming system to support spike sorting with significantly reduced bandwidth; (3) Neural interference causes significant spike power and feature variations, requiring a streaming system to dynamically adapt its operation to maintain robust performance.

4 Design of Neuralite

Motivated by the measurement results in section 3, we propose *Neuralite*, a server-driven brain signal streaming system that significantly reduces bandwidth consumption without compromising spike sorting performance. Our design is independent of spike sorting and downstream neural decoding, allowing it to generalize across a wide range of current and future iBCI applications.

4.1 Overview

At a high-level, the workflow of Neuralite can be divided into three phases. *(1) One-time initialization.* To help the sorting algorithm establish spike templates, the headstage begins by transmitting a small amount of full-resolution signals. This process converges quickly after collecting around 80 spikes per neuron, which requires about 8-min of recording at typical firing rates [3]. Since the initialization does not require real-time streaming, it is not constrained by wireless link capacity and can be implemented with minimal buffering costs. *(2) Training.* After initialization, the streaming controller exports spike templates and firing rates from the sorting algorithm and feeds them to a high-fidelity neural interference simulator. Through extensive profiling of brain signal variability, the controller establishes two empirical models to characterize the stochastic distributions of spike power and features. Based on the power model, the controller identifies a core subset of electrodes that can reliably detect the spikes from all neurons, which is then sent to the headstage to initiate streaming. *(3) Server-driven streaming.* For each brain signal segment, the headstage first sends a basic frame recorded by the core electrode subset. The controller analyzes the basic frame and provides feedback, guiding the headstage in transmitting an enhancement frame that contains key spike features. To minimize bandwidth consumption under dynamic neural interference, the feedback is optimized based on the spike feature model and adapted to the real-time intensity of interference. To track implant drift and MEA property changes, the controller evaluates spike template variation and adaptively triggers retraining when the variation accumulates to a statistically significant level. Note that retraining requires only updated spike templates and does not need a retransmission of full-resolution signals. As a result, the incurred communication overhead is negligible.

4.2 Modeling neural interference

Due to dynamic neural interference, spike power and features can fluctuate significantly, requiring careful modeling to account for this variability. Ideally, these models should be directly profiled from full-resolution signals during the initialization process. However, because spike firing is highly stochastic, direct profiling would either result in prohibitive communication overhead or poor fidelity due to inadequate sampling of interference conditions. To address this, we propose to profile brain signal variability using a probabilistic modeling framework that can faithfully simulate interference scenarios. We build this simulation environment by incorporating spike templates and firing rates into a classic Poisson spike train model [48], and then sample the Poisson process extensively using a Monte Carlo method. This approach allows our streaming controller to "experience" extensive interference conditions with only a small amount of full-resolution signals.

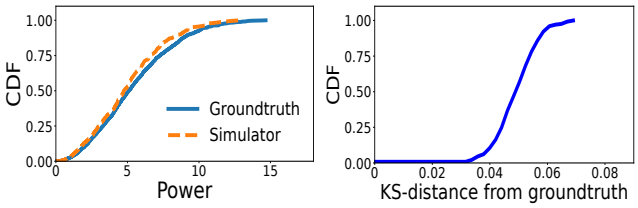

Figure 8: Left: Comparison of spike power distributions profiled by the interference simulator versus the groundtruth measured over an extended period. Right: the KS-distance between simulated power distribution and the groundtruth across all electrodes.

Specifically, for each neuron n, we model the stochastic interference as a probabilistic superimposition of spike templates, expressed as,

$$I_n = \sum_i P(f_{i,n}) A_i(\Delta t_i). \tag{1}$$

where A_i represents the spike template of a potentially interfering neuron i; $f_{i,n}$ denotes the correlated firing rate between neuron i and n, which is measured during the initialization process; $P(f_{i,n})$ is the Poisson function indicating whether i is firing simultaneously with n; and Δt_i represents the random firing timing of the interfering spike. By repeatedly sampling I_n, we can generate a series of *interference templates*, each comprising a matrix that represents the superimposed waveforms of simultaneously firing neurons. Collectively, these interference templates form an *interference tensor*, which we use to model spike power and feature distributions, as described in section 4.3 and section 4.4. To validate our method, we applied it to the Human-1 dataset and compared the spike power distributions profiled uisng our interference tensors with those directly measured using full-resolution recordings over a long period. As shown in Fig. 8, the distribution profiled by our method closely matches the groundtruth measured over extended periods.

In practice, spike templates may vary over time due to implant drift and gradual changes in the MEA's electrical properties. As a result, the interference tensor must be updated accordingly. However, accurately triggering this update is

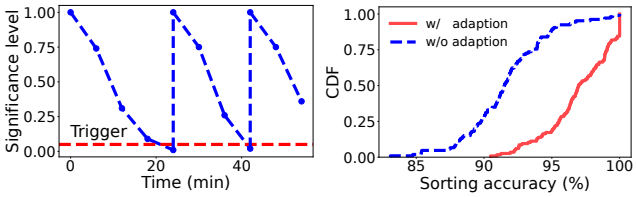

Figure 9: Traces of spike peak divergence (left) and the impact of retraining on sorting accuracy (right).

challenging due to noisy variations in spike power measurements. To address this issue, we propose a metric called*spike peak divergence*, which enables the streaming controller to accurately detect the spike template variation despite significant noise in individual spike measurements. Specifically, we treat the peak powers of spike templates as a probability distribution and use KL divergence to characterize the collective variation in spike templates across the entire neuron population. Each time the sorting algorithm updates spike templates, we compute the spike peak divergence and perform a hypothesis test to assess the statistical significance of the variation. Fig. 9 shows the trace of spike peak divergence during a 54-minuet brain recording using the 374-electrode Neuropixel probe. We observe that adaptation is triggered around every 20 minutes, resulting in a 6.3% improvement in sorting accuracy.

4.3 Optimizing the basic frame

Motivated by the measurement results in section 3, we exploit the widespread distribution of spike power to construct a core subset of electrodes that can detect the spikes from all neurons. We achieve this in two steps.

Electrode selection. To optimize electrode selection, we construct a power graph model similar to the one described in section 3, but augment it to account for dynamic neural interference. Specifically, for each pair of neuron n and electrode e, we profile the spike power of n measured by e using the interference tensor generated in section 4.2. A neuron n is considered as detectable by e if the T-th percentile of spike peak exceeds noise floor by 3 dB, where T is a spike detection threshold preset to 95%. T can also be exposed as a tunable parameter, allowing the neural decoder to balance bandwidth efficiency and spike miss rate according to the application's requirements. Then, we construct an interference-aware power graph by linking electrodes and their detectable neurons. Fig. 10 shows the degrees of electrode vertices across the MEA for two brain signal datasets. We observe that, although many electrodes exhibit high degrees, their spatial distribution is highly random. Motivated by this observation, we perform a greedy search to select electrodes until all neuron vertices are covered. Across the datasets studied, we found that the selected electrode subset is typically 4x to 10x smaller than the MEA's electrode count.

Sample rate optimization. Once the core electrode subset is determined, we further optimize the sample rates of selected electrodes to reduce the basic frame size. To achieve this, we exploit neurons in close proximity to electrodes, which yield high spike power measurements and thus can tolerate certain sampling gaps. We note that sampling with gaps differs from downsampling, where the latter requires anti-aliasing filtering at the headstage and thus adds extra signal processing overhead. In contrast, to reduce sample rate by a factor of N, Neuralite simply divides the sample sequence into segments of N samples and selects only the first sample from each segment, skipping the remaining $N-1$. Given the stochastic nature of spike timing, this sampling method randomly skip parts of a spike. To account for this impact, we run another round of greedy optimization on the selected electrode set. In each iteration, we identify the electrode that can tolerate the maximum sampling gap N while still maintaining coverage for all neurons. Before executing the greedy search, we sort the electrodes in descending order based on their neuron degree. This further reduces the search overhead because electrodes with higher neuron degrees are more likely to be selected for the core subset. In our empirical experiments, we observed that optimizing the sample rates can further reduce the basic frame size by up to 4.2x across the studied datasets.

4.4 Deriving feedback

Upon receiving a basic frame, the streaming controller derives feedback in two steps.

Identifying likely fired neurons. Based on the received basic frame, the streaming controller first identifies a group of likely fired neurons. To this end, a naive method is to apply power thresholding, identify electrodes that detected spike power, and label all their detectable neurons as likely fired. However, this method will lead to a high false detection rate, given the widespread distribution of spike powers. To address this issue, we propose to fuse power detection results across the core electrode subset. Specifically, we label a neuron as likely fired only when all electrodes connected with this neuron in the power graph have detected powers.

Saliency analysis. Given a group of likely fired neurons, the streaming controller constructs an enhancement frame that contains their spike features. To minimize bandwidth consumption, we exploit the relatively concentrated distribution of spike features to reduce the size of enhancement frames. To achieve this, we estimate spike feature distributions through *saliency* analysis. In computer vision, saliency is widely used to explain the significance of pixels to neural network output [21]. By applying saliency analysis, our streaming controller can identify key regions in the full-resolution frame that have critical influence on spike sorting

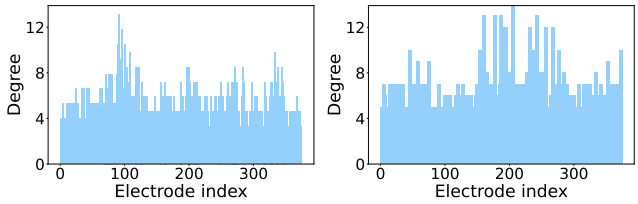

Figure 10: An example of wide distribution of neuron power in a human brain dataset (left) and rat dataset (right).

performance. Fig. 11 shows one example of a saliency map for a spike. We observe that selecting the top 1.1% sample positions with the highest saliency values (as highlighted in the figure) is sufficient for accurately sorting the spike. However, unlike image and video objects, spikes in HR brain signals are subject to extensive interference from other simultaneously fired neurons. As shown in the left figure of Fig. 11, when a spike overlaps with another spike, the resulting interference forces the sorting algorithm to examine 1.5x more sample positions to achieve the same level of sorting accuracy. Therefore, saliency-based feedback must be carefully adapted to account for neural interference.

Adaptation to interference. Instead of selecting a fixed number of sample positions to construct the enhancement frame, Neuralite adapts the number of selected positions based on real-time interference intensity. We leverage the observation that, although neural interference is highly stochastic, its frequency can be estimated based on the firing rates of nearby neurons. Specifically, for each neuron i, we quantify the impact of interference as $\mathcal{I}_i = \sum f_{i,j} e_{i,j}$, where $f_{i,j}$ is the correlated firing rate of neuron j measured in the initialization process, and $e_{i,j}$ is the number of electrodes that can detect the powers of both neurons. Essentially, $\mathcal{I}_i$ computes the expected number of corrupted sample positions when the spikes from two neurons collide. Using this interference measure, we develop an empirical saliency-based feature model to map $\mathcal{I}_i$ to an optimal sample positions, S. We profile this model with a one-time effort using the interference tensors generated in section 4.2.

5 Headstage Implementation

Transmission pipeline. We implement Neuralite headstage based on a dual-core low-power MCU [66]. A straightforward method, as illustrated in Fig. 12(a), is to implement server-driven streaming in sequential steps. However, due to the server feedback latency and the low memory speed of resource-constrained headstages, this method would significantly reduce bandwidth utilization. Our measurement shows that, on a typical low-power MCU, frame construction and the server feedback latency can consume up to 11.1% and 82.3% of processing time, respectively, severely limiting the efficiency of server-driven streaming.

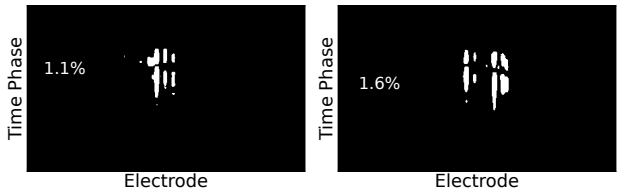

Figure 11: An example of a spike saliency map with and without neural interference.

To address this issue, we employ a pipelined transmission scheduler that processes frames in batches, as depicted in Fig. 12(a). Leveraging the dual-core architecture of modern low-power MCUs, we orchestrate the dual core for parallel frame construction and transmission. Moreover, to make efficient use of the server feedback latency, our headstage continues constructing and transmitting basic frames until it receives the feedback for the first basic frame in the batch. On our headstage prototype, we observe that grouping 8 frames into a batch improves bandwidth utilization by 4.6x.

Buffer control. Our pipelined transmission scheduler requires the headstage to buffer an entire batch of brain signals. Given the high data rate of HR brain signals and the limited memory resources on MCUs, this buffering cost can become particularly significant. To address this challenge, we exploit the fact that the spike firing process is inherently bursty [48], as illustrated in Fig. 12(b). Therefore, instead of buffering an entire brain signal segments, Neuralite buffers signals only from electrodes that are actively recording neural activities. Specifically, we buffer signals from an electrode if it was selected for constructing the basic or enhancement frames within a time window t. Empirically, we found that setting t to 30 ms achieves a good balance between memory efficiency and spike miss rate. For instance, Fig. 12(c) shows that when t increases from 10 ms to 30 ms, the spike miss rate decreases from 5.1% to 0.9%. Beyond 30 ms, further increasing T results in only a 0.3% reduction in miss rate, while memory cost increases by 14.6%. Across the datasets studied, we observed that our buffer controller can reduce buffering cost by up to 34.7%, while limiting spike miss rate below 0.9%.

Hardware design. We prototype our headstage on a custom, coin-size PCB, as shown in Fig. 12(d). Our prototype measures 26.9mm × 21mm × 2mm in size and weighs 4.64g excluding battery, making it comparable to the most compact commodity devices and research prototypes [4, 36–39]. Our headstage hardware hosts ESP32 [66], a low-power system-on-chip MCU integrated with 802.11n. The ESP32 is equipped with a dual-core 240 MHz 32-bit CPU, which consumes about 600 mW in active Wi-Fi transmission mode. The available RAM consists of 320 KB DRAM and 200 KB IRAM, with approximately 362 KB allocated for the OS kernel and TCP/IP stack. To fully utilize the remaining memory, we customize our buffer controller to make use of IRAM by reading and

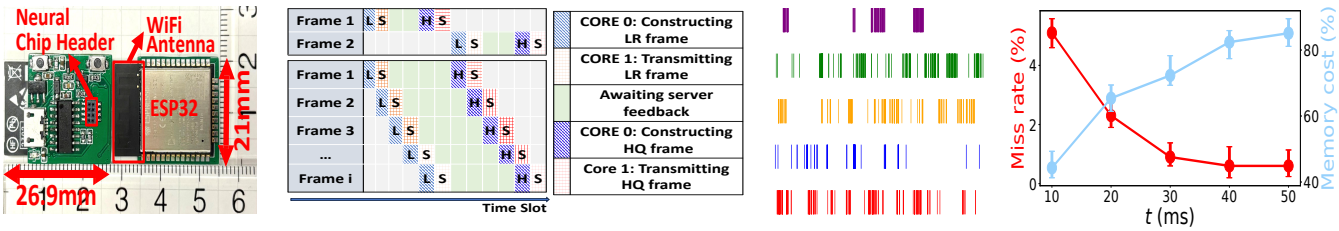

Figure 12: From left to right: (a) headstage prototype; (b) sequential serve-driven streaming (top) versus pipelined transmission of Neuralite (bottom); (c) bursty spike activity of 5 neurons over 180s; (d) memory cost versus spike miss rate as a function of window size.

writing brain signals in 32-bit aligned manner. Our prototype receives digitized HR brain signals via an SPI header, and can be directly connected to commodity neural analog front ends (AFE), such as Intan RHD [22]. Due to the low clock rate of ESP32, the maximum SPI speed of our prototype is limited to 80 Mbps. This limitation creates a bottleneck, restricting our headstage to support at most 400 electrodes at a sampling frequency of 20 kHz with 10 bits per sample. This bottleneck could be overcome by incorporating more powerful MCUs capable of faster SPI communication, which we leave for future work.

6 Evaluation

Neuralite is implemented as an application-layer protocol over TCP. We deploy the streaming controller on a desktop equipped with an 8-Core 5.2 GHz CPU and optimize its performance extensively using OpenCV [41] and OpenMP [42]. When processing brain signals recorded by a 374-electrode MEA at 400 fps, our controller can generate real-time feedback within 0.71 ms per frame. We evaluate Neuralite's performance in supporting two popular spike sorting algorithms [44, 77] and four representative neural decoders [13, 56, 60, 73]. All experiments are conducted in an office environment with extensive Wi-Fi interference from a large-scale campus network.

6.1 Methodology

Experiment setup. While the compact form factor of our headstage is readily fit for in-vivo experiments on small animals, for ethical considerations, we evaluate Neuralite by replaying brain signal datasets recorded using three representative types of MEAs. The recording sites of these datasets span major brain regions in human, macaque, mice, and rats, as summarized in reftab:datasets. The headstage reads the datasets at the designated sample rate from an SD card via SPI. Since the data rates of the Neuropixel datasets exceed the maximum SPI speed of our headstage prototype, we downsampled these datasets from 30 kHz to 20 kHz and reduced the bit quantization to 10 bits per sample. During replay, if the headstage buffer becomes full (e.g., due to slow processing or transmission speed), incoming frames are discarded to prevent overflow.

Baselines. We compare Neuralite against seven representative baselines. For a fair comparison, we implement these baselines on our dual-core MCU, allowing them to construct and transmit frames in a parallel manner similar to Neuralite.

- FullRes streams signals at full resolution. Unlike Neuralite, FullRes does require signal buffering. For a fair comparison, we allocate additional RAM to extend FullRes' TCP sender buffer, which leads to approximately 1.33x increase in throughput.
- NaiveNL is a straightforward implementation of server-driven streaming, which sends base and enhancement frames sequentially without using the transmission pipeline proposed in section 5.
- PwrDet [15] is a headstage-driven streaming scheme. Instead of sending all signals, PwrDet transmits only spike waveforms. To achieve this, it monitors signal power on each electrode and streams a 2 ms segment (the typical spike length) around each detected power peak.
- DCT [59] and FWT [20] compress signals using Discrete Wavelet Transformation and Fast Walsh Transform, respectively. We configure their compression ratios to 15% and 25%, as these settings achieved the best balance between bandwidth efficiency and sorting accuracy in our empirical experiments.
- ASIC DCT and ASIC FWT use ASIC to compress signals, allowing fast compression without adding extra processing burden on MCUs. However, integrating a compression IC to the headstage would incur significant hardware cost. Instead, we simulate a compression IC by directly feeding compressed signals into our headstage. Based on empirical experiments, we set the compression ratio for both ASIC DCT and ASIC FWT to 25%.

6.2 Spike sorting performance

To evaluate the impact of different streaming schemes on spike sorting performance, we measure the firing rate (FR) error, which is computed as the ratio of missing spikes averaged

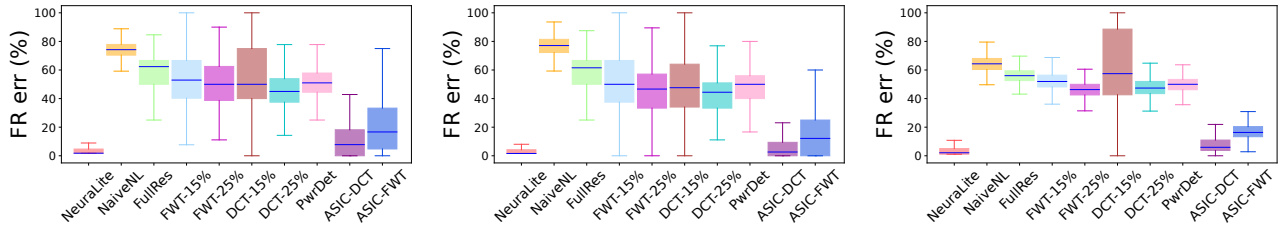

Figure 13: FR errors of Kilosort3 (left and middle) and SpyKING (right) on the datasets recorded using Neuropixel probes.

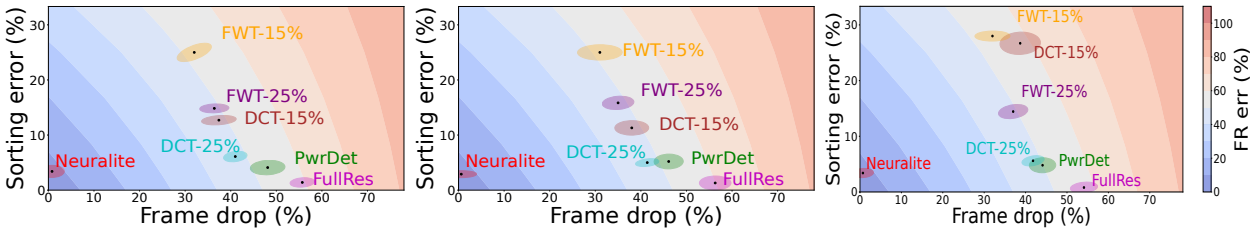

Figure 14: Error break down of the FR errors shown in Fig. 13.

across the entire recorded neuron population. To compute this ratio, we use the off-line sorting results of full-resolution signals as the groundtruth.

Overall FR error. We first study FR error rates on the three most challenging datasets recorded using 374-electrode Neuropixel probes. In this experiment, the headstage is deployed 0.5 m away from the server, where the available TCP throughput is approximately 32.6 Mbps, as measured using iPerf [23]. Fig. 13 compares the overall FR error rates of different streaming schemes. Neuralite consistently outperforms all baselines across the three datasets, limiting the average FR error to between 3.1% and 4.3%. As expected, ASIC DCT and ASIC FWT perform better than other baselines; however, their average FR error rates are still 2.4x and 3.1x higher than Neuralite's, primarily because they compress brain signals blindly without effectively preserving spike features. We also observe that NaiveNL exhibits the worst FR error rate among all baselines. This poor performance is mainly caused by server feedback latency and limited memory speed, which severely degrades NaiveNL's bandwidth utilization. While both Neuralite and NaiveNL follows the server-driven streaming workflow, the sharp contrast in their performance highlights the effectiveness of Neuralite's pipelined transmission scheme proposed in section 5.

FR error breakdown. To further analyze the causes of FR errors, we break down the FR errors of Neuralite and baselines into two components: the *frame drop rate* at the headstage and the *sorting error rate* at the server. The frame drop rate reflects the impacts of processing and communication burden on the headstage, which causes incoming frames to be discarded before entering the buffer. The sorting error rate is calculated as the ratio of missing spikes within the frames received by the server, which reflects the impacts of spike waveform distortion caused by streaming schemes.

Fig. 14 shows the breakdown of the FR errors measured in the previous experiment. For ASIC-based compression schemes, we observe that their frame drop rate decreases rapidly as the compression ratio increases. This is expected because aggressive compression reduces bandwidth consumption, allowing the headstage to transmit more frames. However, the overall spike sorting performance does not show a corresponding improvement, mainly due to the increased sorting error rate caused by compression-induced spike distortions. Unlike ASIC-based compression, PwrDet maintains a low sorting error rate but suffers from significantly higher frame drops. This is caused by two factors. First, PwerDet requires the headstage to detect and extract spikes, introducing substantial computational overhead and thus reducing the processing throughput. Second, compared to compression, PwrDet is less bandwidth efficient, leading to a reduced frame rate. This is because the spike power of HR brain signals is often widely distributed across electrodes. By simply streaming signals based on power detection, PwrDet can cause considerable bandwidth waste. In comparison, Neuralite is able to effectively balance bandwidth efficiency and sorting accuracy. Compared to PwerDet, Neuralite is able to reduce frame drop rate by 28.2x to 51.7x, while keeping the sorting error rate to below 3.4%, which is 1.8x better than that of compression-based schemes.

Neuron-wise FR error distribution. We then analyze the FR error for individual neurons. To filter out outliers, we exclude neurons that fire less than 0.2 times per second. As shown in Fig. 15, Neuralite limits FR error to 0% for more than 50% of neurons. Moreover, it keeps the 90% percentile FR error rate below 10%, which is 1.3x, 6.9x, and 8.2x better compared to ASIC-DCT, DCT, and PwrDet, respectively. These results demonstrate Neuralite's superior ability to preserve

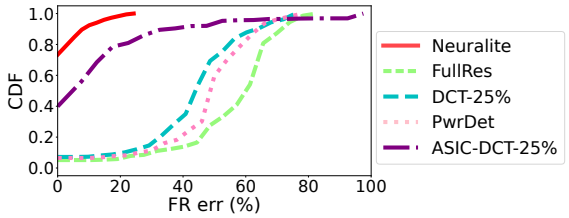

Figure 15: Neuron-wise FR error variations across recorded neuronal populations.

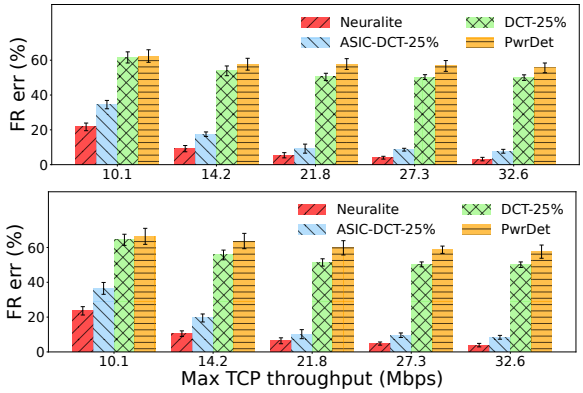

Figure 16: FR error rates as a function of available TCP throughput on the Human (top) and Rat (bottom) datasets recorded by Neuropixel probes.

sorting accuracy for the majority of neurons in large-scale brain activity recordings.

6.3 Benchmarks

Next, we conduct a series of benchmarks to evaluate Neuralite under various wireless bandwidth, neural activity intensities, MEA sizes, and MEA densities.

Available bandwidth. We first compare Neuralite with other streaming schemes under different wireless bandwidth conditions. To control the available bandwidth, we use clumsy [8] to limit the maximum TCP throughput at the server. Fig. 16 plots the FR errors as a function of TCP throughput for the Human-1 and Rat datasets. We observe that, in both experiments, Neuralite outperforms all baselines across all throughput conditions. For example, to maintain FR errors below 5% for the Rat dataset, Neuralite requires a TCP throughput of 21.8 Mbps and 22.6 Mbps, respectively. In comparison, the simulated ASIC-DCT requires approximately 28.5 Mbps, which is 1.3x higher.

Neural activity. To evaluate Neuralite's performance under different levels of neural activity, we use eMouse [30], a popular neural activity simulator widely used by the neuroscience community, to synthesize HR brain signals. Specifically, we generate simulated neurons by inputting the spike templates measured in the Human dataset into eMouse, and adjust the firing rates of the simulated neurons in proportion to the firing rates of real neurons observed in the original datasets. We then synthesize neural activity using eMouse and play the generated signals using our headstage. As shown in Fig. 17, the FR errors for both Neuralite and PwrDet increase as the firing rates of simulated neurons increase. However, PwrDet is far more sensitive than Neuralite, as its processing burden and bandwidth efficiency rapidly degrade with increased neural activity. In contrast, Neuralite maintains more stable performance due to its adaptive streaming framework.

MEA size. To analyze the impact of MEA size, we evaluate Neuralite's performance on the Macaque, Mice, and Human datasets recorded using MEAs of 96, 256, and 374 electrodes, respectively. Fig. 18 (left) compares the FR error rates of Nerualite, PwerDet, and DCT. We observe that all streaming schemes maintain low FR error rates on the Macaque dataset, due to the low bandwidth requirement. However, as MEA size increases, PwrDet and DCT both experience significant increases in FR error rates. In contrast, Neuralite limits the FR error rate to below 5%. To further assess the impact of MEA size, we evaluate the minimum TCP throughput required to support MEAs of different sizes. For this experiment, we use the Human datasets recorded with a 374-electrode Neuropixel and replay signals from a selected continuous subset of electrodes. We then gradually increase available TCP throughput using clumsy until the FR error rate drops to below 5%. As shown in Fig. 18 (right), DCT and PwrDet can support at most 165 and 220 electrodes, respectively. In comparison, Neuralite is able to support all 374 electrodes.

Neuron density. The density of recorded neurons is a key factor influencing the distribution of spike power and the intensity of neural interference. To analyze this impact, we compare the performance of Neuralite, PwrDet and DCT on the Macaque, Mice, and Human datasets, which are recorded using MEAs of different electrode densities. To eliminate the influence of neuron count, we trimmed the Mice and Human datasets by selecting signals from a continuous subset of electrodes, ensuring they contain a similar number of neurons as the Macaque dataset. To scale the transmission rate with the reduced neuron count, we limit the maximum TCP throughput to 4.5 Mbps. To measure neuron density, we calculate the average degrees of neuron vertices in the power graphs.

As shown in Fig. 19 (left), as neuron density increases, the FR error of PwrDet and DCT rises consistently, primarily due to their susceptibility to the increased intensity of neural interference. In contrast, when neuron density increases from 1.5 to 3.1, Neuralite's FR error decreases from 10% to 4%. This improvement is due to the fact that, as neuron density increases, each electrode records a larger number of neurons, allowing Neuralite to construct a smaller core

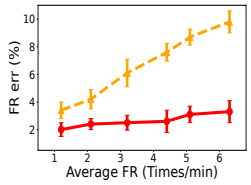

Figure 17: The impact of neural activity on FR error rate.

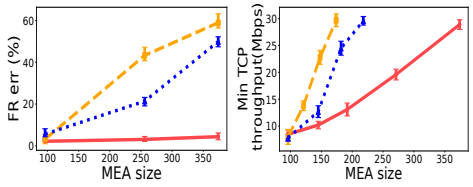

Figure 18: The impact of MEA size (left) and the minimum TCP throughput required to support a MEA of a specific size (right).

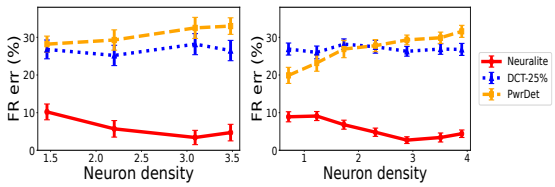

Figure 19: Impact of MEA density on FR error rate.

electrode subset to cover all neurons, thereby reducing bandwidth consumption. However, when neuron density further increases to 3.5, the commensurate rise in neural interference intensity causes Neuralite's FR error to increase from 3% to 5%, but it is still 7.1x and 6x lower than that of PwrDet and DCT, respectively. To further analyze the impact of neuron density, we synthesize brain signals using eMouse based on the Human datasets and adjust the density of active neurons. Fig. 19 (right) shows the FR errors of Neuralite, PwrDet, and DCT across seven synthesized datasets. Consistent with the trend in Fig. 19 (left), Neuralite outperforms PwrDet and DCT consistently and significantly across different neuron densities.

6.4 Impacts on iBCI applications

Sorting errors introduced by the streaming scheme can propagate through the neural decoding pipeline. In the following, we evaluate this impact on four representative iBCI applications.

Methodology. Application-level iBCI datasets typically comprise only post-sorting spike firing traces without the raw spike waveform. This limitation prevents us from directly assessing how different streaming schemes distort spike waveforms. To address this issue, we first obtain neuron firing traces with annotated sorting errors from the raw dataset experiments described in section 6.2, and then inject these error traces into the application datasets. We begin by pairing neurons in the raw and application datasets based on their firing patterns. For each pair, we apply the spike sorting errors measured in the raw dataset experiments into the firing traces of the application dataset. Finally, we re-run the neural decoder of the application to evaluate the impact on application-level performance.

Vision iBCIs Vision iBCIs aim to map brain signals to images, enabling translational tools that can restore visual functions to visually impaired individuals. In this experiment, we study two representative vision iBCIs.

The vision iBCI proposed in [60] utilizes a Convolutional Neural Network (CNN) to translate neural activity into visual stimuli. The dataset is captured from a mouse's visual cortex using a 374-electrode Neuropixel probe. The neural decoder employs Kilosort to track the firing rates of 93 neurons. During training, the mouse is exposed to a set of 100 natural and movie scenes. The CNN then learns to map neuron firing patterns to pixel values for each location in the visual field. In the testing phase, the mouse is shown a new visual stimulus and the vision iBCI records the corresponding neuron firing patterns, which are translated into an image composed of predicted pixel values. Fig. 20 shows the predicted images after injecting the firing rate errors caused by different streaming schemes. Neuralite limits the pixel prediction error rate to below 1.7%. In comparison, PwrDet and DCT result in error rates of 34.8% and 23.9%, respectively.

CEBRA [56] analyzes neural activity to predict the visual scene observed by a mouse. The dataset is recorded using Neuropixels from a mouse's visual cortex and then spike sorted using Kilosort. During training, the mouse is shown a 30s movie at 30 fps. Then, CEBRA embeds the neuron firing traces and trains a kNN decoder to predict which scene the mouse is viewing. To inject firing rate errors, we use the result of the Human dataset experiment. As shown in Fig. 21, Neuralite introduces a minimal increase in prediction error, from 19.4% to 20.7%, which is 19.8x and 13.3x lower than the error increases caused by PwrDet and DCT, respectively.

Motor iBCIs Motor iBCIs translate neural signals into motion intentions. We examine a motor iBCI that predicts a rat's movement trajectory by using CEBRA's kNN [56] to decode neural signals recorded from the rat's hippocampus [13]. Fig. 22 shows the predicted trajectories of the mouse over a 5s interval, after injecting the firing errors caused by Nerualite, PwrDet, and DCT. We observe that Neuralite is able to limit the median trajectory error to within 0.02 m. This increase in trajectory error is 21x and 17x lower than that caused by PwrDet and DCT, respectively.

Brain-to-text communication. The brain-to-text communication system [73] employs a DNN to translate neural activity into attempted handwriting motions. The dataset is recorded in the motor cortex of a paralyzed patient using a 192-electrode Utah array. The neural decoder divides signals into 10 ms bins, counts spikes on each microelectrode, and takes the measured spike rate vector as input. To evaluate the impacts of different streaming schemes on this neural

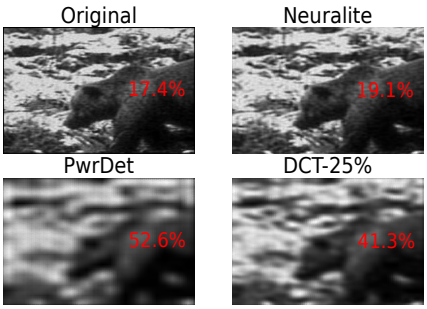

Figure 20: Visual stimulus recovery.

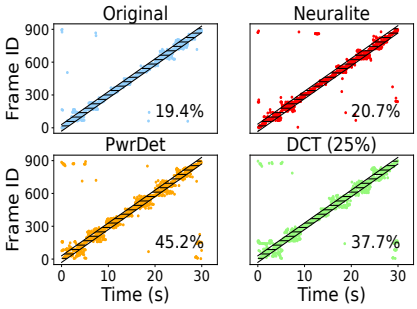

Figure 21: Visual stimulus prediction.

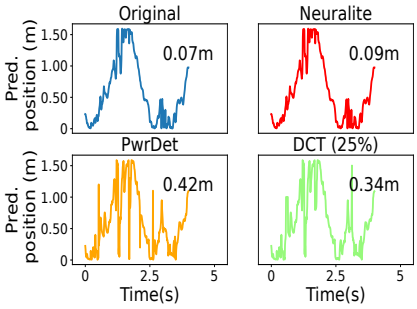

Figure 22: Impacts on a motor iBCI.

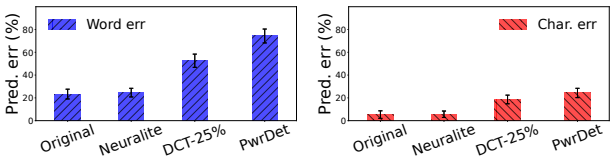

Figure 23: Impact on handwriting application.

decoder, we measure the corresponding spike rate vectors in the raw dataset experiments and then inject the spike sorting error trace. We observe that Neuralite slightly increases the character error rate from 5.3% to 5.7%. In contrast, PwrDet and DCT lead to significantly higher character error rates of 24.3% and 18.6%, respectively, severely degrading text accuracy. Character errors, in turn, increase the likelihood of word errors. Neuralite keeps the word error increase to just 1.5%, which is 19.5x and 31.1x lower than that of DCT and PwrDet, respectively

7 Discussions

Impact of RF interference. Like all wireless systems, wireless iBCIs are susceptible to RF interference and malicious jamming. However, although the current implementation of Neuralite uses Wi-Fi and may be by ambient interference, its design is independent of specific wireless technology. This flexibility allows Neuralite to integrate frequency hopping and customized radios operating in less crowded bands to improve interference resilience.

Application-aware optimization. The current design of Neuralite focuses on optimizing brain signal streaming for spike sorting, which ensures its generalizability and compatibility with a wide range of downstream iBCI applications. However, Neuralite does not consider application-layer neuron functions, where the significance of neurons may vary greatly across the recorded population. As a result, applying the same sorting error constraint to all neurons may result in sub-optimal performance. To address this issue, a potential solution is to leverage Explainable AI (XAI) to gain insights into neural decoders and optimize sorting error constraints for individual neurons based on their specific contributions to neural decoding performance.

Adaptation to bandwidth variation. Wireless link capacity can fluctuate with time-varying interference and channel conditions. While the current design of Neuralite focuses on minimizing bandwidth consumption, it does not account for the impact of bandwidth variations. To address this issue, a potential solution is to dynamically adjust Neuralite's parameters based on the available bandwidth. One example is to adjust the spike detection constraint to balance spike sorting performance with communication efficiency under the constraint of available link capacity. This adaptation strategy will be explored in future work.

8 Conclusion

We presented Neuralite, an efficient brain signal streaming system that enables wireless iBCIs under limited bandwidth. Through high-fidelity brain signal modeling and server-driven streaming optimization, Neuralite effectively addresses key challenges in iBCIs, including severe resource constraints on headstages, the complexity of HR brain signals, and dynamic neural activity. By overcoming these challenges, Neuralite holds potential to support a wide range of important research and clinical applications that require untethered iBCIs, such as Neurophilosophical studies and advanced prosthetics.

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